Set Algebra — Formula Sheet + Practice

Universe: Fix a universe U. All complements are taken in U: $A^c = U \setminus A$.
Key proof method: $A = B \iff (\forall x)\,[x \in A \Leftrightarrow x \in B]$.

Basic Definitions

- Union: $A \cup B = \{x : x \in A \text{ or } x \in B\}$
- Intersection: $A \cap B = \{x : x \in A \text{ and } x \in B\}$
- Difference: $A \setminus B = \{x : x \in A \text{ and } x \notin B\}$
- Complement: $A^c = U \setminus A$
- Symm. difference: $A \triangle B = (A \setminus B) \cup (B \setminus A)$

Algebra Laws (Union/Intersection)

- Commutative:
$$A \cup B = B \cup A,$$
$$A \cap B = B \cap A$$
- Associative:
$$(A \cup B) \cup C = A \cup (B \cup C),$$
$$(A \cap B) \cap C = A \cap (B \cap C)$$
- Distributive:
$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$
$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$
- Idempotent: $A \cup A = A,\ \ A \cap A = A$
- Absorption: $A \cup (A \cap B) = A,\ \ A \cap (A \cup B) = A$
- Identity: $A \cup \varnothing = A,\ \ A \cap U = A$
- Domination: $A \cup U = U,\ \ A \cap \varnothing = \varnothing$

Complement Laws

- Involution: $(A^c)^c = A$
- $A \cup A^c = U,\ \ A \cap A^c = \varnothing$
- De Morgan:
$$(A \cup B)^c = A^c \cap B^c, \qquad (A \cap B)^c = A^c \cup B^c.$$

Difference & Symmetric Difference

- Rewrite: $A \setminus B = A \cap B^c$
- $(A \cup B) \setminus C = (A \setminus C) \cup (B \setminus C)$
- $A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$
- $A \setminus (B \cap C) = (A \setminus B) \cup (A \setminus C)$
- $(A \setminus B)^c = A^c \cup B$
- $A \triangle B = (A \cup B) \setminus (A \cap B)$
- $A \triangle A = \varnothing,\ \ A \triangle \varnothing = A,\ \ A \triangle U = A^c$

Membership Translation (Logic Dictionary) For any element x:

$$x \in A \cup B \iff (x \in A) \vee (x \in B)$$
$$x \in A \cap B \iff (x \in A) \wedge (x \in B)$$
$$x \in A^c \iff \neg(x \in A)$$
$$x \in A \setminus B \iff (x \in A) \wedge \neg(x \in B)$$
$$x \in A \triangle B \iff \text{“exactly one of } x \in A, x \in B\text{” (XOR)}$$

Power Set Properties (for $\mathcal{P}(A)$)

- $A \subseteq B \iff \mathcal{P}(A) \subseteq \mathcal{P}(B)$.
- If A is finite, then $|\mathcal{P}(A)| = 2^{|A|}$.

Cardinality / counting identities (Finite Sets)

- **Two sets (Inclusion–Exclusion):**
$$|A \cup B| \ = \ |A| + |B| - |A \cap B|.$$
- **Disjoint Union Special Case:** If $A \cap B = \varnothing$, then
$$|A \cup B| \ = \ |A| + |B|.$$
- **Three sets (Inclusion–Exclusion):**
$$|A \cup B \cup C|$$
$$= \ |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|.$$
- **Difference:**
$$|A \setminus B| \ = \ |A| - |A \cap B|.$$
- **Complement (Relative to U):**
$$|A^c| \ = \ |U| - |A|.$$
- **Symmetric Difference:**
$$|A \triangle B| \ = \ |A \setminus B| + |B \setminus A| \ = \ |A| + |B| - 2|A \cap B|.$$

Cartesian Product Properties

- **Distributive Over Union/Intersection:**
$$A \times (B \cup C) \ = \ (A \times B) \cup (A \times C),$$
$$A \times (B \cap C) \ = \ (A \times B) \cap (A \times C).$$
- **Distributive over union/intersection (left factor):**
$$(A \cup B) \times C \ = \ (A \times C) \cup (B \times C),$$
$$(A \cap B) \times C \ = \ (A \times C) \cap (B \times C).$$
- **Difference:**
$$(A \setminus B) \times C \ = \ (A \times C) \setminus (B \times C),$$
$$A \times (B \setminus C) \ = \ (A \times B) \setminus (A \times C).$$
- **Empty set:**
$$A \times \varnothing \ = \ \varnothing, \qquad \varnothing \times A \ = \ \varnothing.$$
- **Cardinality (finite sets):**
$$|A \times B| \ = \ |A|\,|B|.$$

1 Practice Identities with Solutions

1. **Identity elements.** Prove $A \cup \varnothing = A$ and $A \cap U = A$.
   Solution. For any x:

$$x \in A \cup \varnothing \iff (x \in A) \vee (x \in \varnothing) \iff (x \in A) \vee (\text{false}) \iff x \in A.$$

   So $A \cup \varnothing = A$. Similarly,

$$x \in A \cap U \iff (x \in A) \wedge (x \in U) \iff (x \in A) \wedge (\text{true}) \iff x \in A,$$

   so $A \cap U = A$.

2. **Difference as intersection with complement.** Prove $A \setminus B = A \cap B^c$.
   Solution. For any x:

$$x \in A \setminus B \iff (x \in A) \wedge (x \notin B) \iff (x \in A) \wedge (x \in B^c) \iff x \in A \cap B^c.$$

3. **De Morgan (union).** Prove $(A \cup B)^c = A^c \cap B^c$.
   Solution. For any x:

$$\begin{aligned} x \in (A \cup B)^c &\iff x \notin (A \cup B) \iff \neg\big((x \in A) \vee (x \in B)\big) \\ &\iff \big(\neg(x \in A)\big) \wedge \big(\neg(x \in B)\big) \iff (x \in A^c) \wedge (x \in B^c) \iff x \in A^c \cap B^c. \end{aligned}$$

4. **Distributive law.** Prove $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.
   Solution. For any x:

$$\begin{aligned} x \in A \cap (B \cup C) &\iff (x \in A) \wedge \big((x \in B) \vee (x \in C)\big) \\ &\iff \big((x \in A) \wedge (x \in B)\big) \ \vee \ \big((x \in A) \wedge (x \in C)\big) \\ &\iff x \in (A \cap B) \cup (A \cap C). \end{aligned}$$

5. **Absorption.** Prove $A \cup (A \cap B) = A$.
   Solution. For any x:

$$x \in A \cup (A \cap B) \iff (x \in A) \vee \big((x \in A) \wedge (x \in B)\big).$$

   In logic, $P \vee (P \wedge Q) \iff P$. Taking $P = (x \in A)$ gives $x \in A \cup (A \cap B) \iff x \in A$.

6. **Difference of a union.** Prove $A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$.
   Solution. For any x:

$$\begin{aligned} x \in A \setminus (B \cup C) &\iff (x \in A) \wedge \neg\big((x \in B) \vee (x \in C)\big) \\ &\iff (x \in A) \wedge \big(\neg(x \in B) \wedge \neg(x \in C)\big) \\ &\iff \big((x \in A) \wedge \neg(x \in B)\big) \wedge \big((x \in A) \wedge \neg(x \in C)\big) \\ &\iff x \in (A \setminus B) \cap (A \setminus C). \end{aligned}$$

7. **Complement of a difference.** Prove $(A \setminus B)^c = A^c \cup B$.
   Solution. Using $A \setminus B = A \cap B^c$:

$$(A \setminus B)^c = (A \cap B^c)^c.$$

   By De Morgan (intersection),

$$(A \cap B^c)^c = A^c \cup (B^c)^c = A^c \cup B.$$

8. **Subtracting twice.** Prove $(A \setminus B) \setminus C = A \setminus (B \cup C)$.
   Solution. For any x:

$$\begin{aligned} x \in (A \setminus B) \setminus C &\iff x \in A \setminus B \text{ and } x \notin C \\ \iff \big((x \in A) \wedge (x \notin B)\big) \wedge (x \notin C) &\iff (x \in A) \wedge \big((x \notin B) \wedge (x \notin C)\big) \\ \iff (x \in A) \wedge \neg\big((x \in B) \vee (x \in C)\big) &\iff x \in A \setminus (B \cup C). \end{aligned}$$

9. **Symmetric difference identity.** Prove

$$A \triangle B = (A \cup B) \setminus (A \cap B).$$

Solution. Start from the definition:

$$A \triangle B = (A \setminus B) \cup (B \setminus A).$$

Rewrite differences:

$$(A \setminus B) \cup (B \setminus A) = (A \cap B^c) \cup (B \cap A^c).$$

Now note: "exactly one of A, B" is the same as "in $A \cup B$ but not in $A \cap B$". Formally, for any x:

$$x \in (A \cap B^c) \cup (B \cap A^c) \iff \big((x \in A) \wedge (x \notin B)\big) \vee \big((x \in B) \wedge (x \notin A)\big)$$

$$\iff \big((x \in A) \vee (x \in B)\big) \wedge \neg\big((x \in A) \wedge (x \in B)\big) \iff x \in (A \cup B) \setminus (A \cap B).$$

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2 More Set Theory Properties

Standing assumptions: $A, B, C \subseteq U$, complements in U.

A. Subset criteria (very common in proofs)

Claim A1. $A \subseteq B \iff A \cap B = A$.
Proof. ($\Rightarrow$) If $A \subseteq B$, then every $x \in A$ lies in B, so $x \in A \cap B$. Hence $A \subseteq A \cap B$. Also $A \cap B \subseteq A$ always, so $A \cap B = A$.
($\Leftarrow$) If $A \cap B = A$, then $A = A \cap B \subseteq B$, hence $A \subseteq B$. $\square$
Claim A2. $A \subseteq B \iff A \cup B = B$.
Proof. ($\Rightarrow$) If $A \subseteq B$, then $A \cup B \subseteq B$ and also $B \subseteq A \cup B$, so equality holds. ($\Leftarrow$) If $A \cup B = B$, then $A \subseteq A \cup B = B$, so $A \subseteq B$. $\square$

B. More difference identities

Claim B1. $(A \cup B) \setminus C = (A \setminus C) \cup (B \setminus C)$.
Proof. For any x,

$$x \in (A \cup B) \setminus C \iff \big((x \in A) \vee (x \in B)\big) \wedge \neg(x \in C)$$

$$\iff \big((x \in A) \wedge \neg(x \in C)\big) \ \vee \ \big((x \in B) \wedge \neg(x \in C)\big) \iff x \in (A \setminus C) \cup (B \setminus C).$$

$\square$

Claim B2. $(A \cap B) \setminus C = (A \setminus C) \cap (B \setminus C)$.
Proof. For any x,

$$x \in (A \cap B) \setminus C \iff (x \in A) \wedge (x \in B) \wedge \neg(x \in C)$$

$$\iff \big((x \in A) \wedge \neg(x \in C)\big) \wedge \big((x \in B) \wedge \neg(x \in C)\big) \iff x \in (A \setminus C) \cap (B \setminus C).$$

$\square$

Claim B3. $A \setminus (B \setminus C) = (A \setminus B) \cup (A \cap C)$.
Proof. Start with membership:

$$x \in A \setminus (B \setminus C) \iff (x \in A) \wedge \neg\big(x \in B \setminus C\big).$$

But $x \in B \setminus C \iff (x \in B) \wedge \neg(x \in C)$, so

$$\neg\big((x \in B) \wedge \neg(x \in C)\big) \iff \neg(x \in B) \ \vee \ (x \in C).$$

Hence

$$x \in A \setminus (B \setminus C) \iff (x \in A) \wedge \big(\neg(x \in B) \vee (x \in C)\big)$$

$$\iff \big((x \in A) \wedge \neg(x \in B)\big) \ \vee \ \big((x \in A) \wedge (x \in C)\big) \iff x \in (A \setminus B) \cup (A \cap C).$$

$\square$

C. Partition-style identities (very useful for counting)

Claim C1. $A = (A \setminus B) \,\dot\cup\, (A \cap B)$. (Disjoint union.)
Proof. First, $(A \setminus B) \cap (A \cap B) = \varnothing$ because x cannot be simultaneously in B and not in B. Next, for any $x \in A$, either $x \in B$ or $x \notin B$; in the first case $x \in A \cap B$, in the second $x \in A \setminus B$. So $A \subseteq (A \setminus B) \cup (A \cap B)$. The reverse inclusion is clear. $\square$
Corollary (finite sets). If A is finite, then $|A| = |A \setminus B| + |A \cap B|$. $\square$

D. More symmetric-difference properties

Claim D1. $(A\triangle B)^c = (A\cap B)\cup(A^c\cap B^c)$.
Proof. Using $A\triangle B = (A\cap B^c)\cup(B\cap A^c)$,

$$(A\triangle B)^c = (A\cap B^c)\cup(B\cap A^c)^c = (A\cap B^c)^c\cap(B\cap A^c)^c.$$

By De Morgan,

$$(A\cap B^c)^c = A^c\cup B, \qquad (B\cap A^c)^c = B^c\cup A.$$

So

$$(A\triangle B)^c = (A^c\cup B)\cap(B^c\cup A).$$

Distribute:

$$(A^c\cup B)\cap(B^c\cup A) = (A^c\cap B^c)\cup(A^c\cap A)\cup(B\cap B^c)\cup(B\cap A).$$

Middle two terms are $\varnothing$, leaving

$$(A\triangle B)^c = (A^c\cap B^c)\cup(A\cap B).$$

□

Claim D2. $A\cap(B\triangle C) = (A\cap B)\triangle(A\cap C)$.
Proof. Start from $B\triangle C = (B\setminus C)\cup(C\setminus B)$:

$$A\cap(B\triangle C) = A\cap(B\setminus C)\ \cup\ A\cap(C\setminus B).$$

Now $A\cap(B\setminus C) = A\cap(B\cap C^c) = (A\cap B)\cap(A\cap C)^c = (A\cap B)\setminus(A\cap C)$. Similarly $A\cap(C\setminus B) = (A\cap C)\setminus(A\cap B)$. Therefore

$$A\cap(B\triangle C) = \big((A\cap B)\setminus(A\cap C)\big)\cup\big((A\cap C)\setminus(A\cap B)\big) = (A\cap B)\triangle(A\cap C).$$

□
